
INSTRUCTIONS TO THE STUDENTS:-

1. Attempt any one question from each group and one question from remaining
 2. Assume suitable data wherever necessary and state the same clearly.
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Group - I

- Q.1** A) Explain the fundamental steps in Digital Image Processing. Also list different applications of digital image processing **05**
B) Explain image replication process with suitable example? Define image interpolation and Explain different image interpolation techniques? **05**
- Q.2** Explain following geometrical transformations in 2-D with an example. **10**
(i) Translation (ii) Scaling (iii) Rotation (iv) Affine Transform

Group - II

- Q.3** Explain the following terms in detail. **10**
(i) Different pixel neighbourhoods (ii) Distance Metrics
(iii) Types of adjacency (iv) Types of connectivity
- Q.4** Define Unitary matrix and Test the following matrices are unitary matrix or not. **10**

$$(i) A = \frac{1}{2} \begin{bmatrix} \sqrt{3} & 1 \\ -1 & \sqrt{3} \end{bmatrix} \quad (ii) A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & j \\ j & 1 \end{bmatrix} \quad (iii) A = \begin{bmatrix} \sqrt{2} & j \\ -j & \sqrt{2} \end{bmatrix}$$

Group - III

- Q.5** Define spatial processing and explain the following terms. **10**
(i) Contrast stretching (ii) Bit plane slicing (iii) Histogram Equalization
- Q.6** Explain Hadamard and Discrete Cosine Transform (DCT) Transform along with their properties? **10**

Group - IV

Q.7	A) Explain the estimation of degradation model: (i) Observation (ii) Experimental (iii) Mathematical	06
	B) Explain image contrast stretching and thresholding?	04
Q.8	A) Explain basic gray level transformations: (i) Identity, Negative (ii) Log, Inverse-log (iii) n^{th} power, n^{th} root	06
	B) Write down MATLAB program for contrast stretching?	04

BEST OF LUCK
